The Homogeneous Propagation Framework: Retardation, Cumulative Field Structure, and Rotating-Source Response

Abstract

Companion papers: The Homogeneous Propagation Framework: Transport, Local Geometry, and Displacement Defects (DOI: 10.5281/zenodo.19079929); *The Homogeneous Propagation Framework: Weak-Field Observational Tests* (companion paper)

This paper develops the next dynamical step of the displacement framework. The starting point is the postulate that sustained excitation loading sources a displacement of the transport-supporting structure, and that changes in that displacement propagate causally at the medium speed c. In the present weak-limit treatment this source is represented by mass-energy, while the deeper constitutive variable is left to be derived. In the stationary spherically symmetric limit, that postulate yields the equilibrium displacement geometry $D(r, d) = \sqrt{r^2 + d^2}/r$, the cumulative displacement field $\Sigma(r)$, and the correct weak-field Newtonian limit. What has been lacking is not nonperturbative static geometry, but the dynamical law that propagates displacement away from equilibrium and the deeper constitutive derivation of its source from the substrate.

The main dynamical claim is that the retarded displacement map is the primary expression of the causal structure. The field at a point depends on retarded source configuration rather than instantaneous source position, and the retarded integral for the cumulative displacement $\Sigma(\mathbf{x}, t)$ has a local wave-equation equivalent propagating at the medium speed c. The displacement defines an effective coordinate-level geometry whose structure is fixed by the framework's own internal logic: vacuum storage capacity is anchored to the coordinate grid, while transport is displaced relative to that grid. The resulting line element modifies the time and radial components while leaving the angular sector Euclidean, and its weak-field orbit equation recovers the standard perihelion precession. Rotating sources require an additional vector response, and the first non-spherical application developed here is the rotating-source

frame-drag branch, whose current Earth-corrected Gravity Probe B estimate lies near the observed weak-field signal.

The status of the framework is therefore more specific than a blanket weak-field limitation. The static displacement geometry is already nonperturbative, and the observational development carried out so far is mainly in the weak-field regime because that is where present tests lie. The genuinely open problem is the completion of the dynamical law: deriving the substrate-loading variable Q_{occ} associated with sustained excitation closure, deriving the loaded-medium scalar and vector response, and recovering $D(r, d)$ as the stationary solution of that common law. The result is not yet a final closed field theory, but it does define a coherent dynamical program: a two-layer coupled structure, a causal retarded displacement map with its wave-equation equivalent, an effective coordinate geometry recovering the correct orbit structure, and a constitutive route toward a unified scalar and rotating-source response.

1. Foundational Postulate and Scope

This framework begins from a single physical postulate at the effective level: **sustained excitation loading sources a displacement of the transport-supporting structure, and changes in that displacement propagate causally at the medium speed c.** In the present weak-limit treatment, that source appears as mass-energy; the deeper constitutive variable, introduced later as Q_{occ}, remains to be derived. In the layered picture already developed in the companion transport/local-geometry paper, this is a statement about the geometry of the propagation-supporting stage itself, not about an additional force law acting within ordinary matter dynamics. The transport structure carries the propagation condition, and when localized source loading inserts volume into that layer, the resulting geometric change appears observationally as gravity, lensing, time dilation, and related weak-field effects.

A useful structural comparison may be made with Wheeler's familiar summary of general relativity: *matter tells spacetime how to curve, and spacetime tells matter how to move.* The present framework adopts an intentionally similar division of roles, but with a different ontology. Here the relation is instead: **mass tells space how to displace, and displaced space tells mass how to move.** The similarity is only at the level of coupling structure. In both cases, one sector describes how sources modify the stage, while another describes motion within the modified stage. In this framework, however, gravity is not identified with curvature of spacetime itself. It is identified with displacement of the transport-supporting spatial network.

This viewpoint leads naturally to two coupled dynamical layers. The first contains the ordinary dynamics of excitations, matter, and interactions evolving within the transport structure. The second contains the displacement of the transport structure itself. These layers are coupled, but not collapsed into a single equation. At the effective level, mass-energy sources displacement, and the resulting displacement modifies the effective geometry through which motion proceeds. Gravity is therefore not treated here as a conventional force internal to transport-structure dynamics, but as a modification of the stage on which those dynamics unfold. The two-layer architecture also clarifies the role of the present paper: the equation to be developed belongs to the displacement layer, not to the full microphysics of matter.

For a stationary isolated source, the equilibrium form of that displacement is described by

$$D(r, d) = \frac{\sqrt{r^2 + d^2}}{r},$$

with d the displacement scale associated with the source. In the existing development, this form is not introduced as an arbitrary ansatz. It follows from the geometric requirement that when a volume $\frac{4}{3}\pi d^3$ is inserted into the spherically symmetric transport structure, each shell originally at radius r is displaced outward to $r' = \sqrt{r^2 + d^2}$, so that the static displacement factor is simply the ratio r'/r. The postulate is therefore physical, while the static form is geometric.

The weak-field structure is then governed not merely by the local deviation $D - 1$, but by the cumulative displacement

$$\Sigma(r) = \int_r^\infty (D(r') - 1)\, dr',$$

whose weak-field limit gives

$$\Sigma(r) \approx \frac{d^2}{2r}.$$

This quantity carries the required $1/r$ behavior, governs clock-rate and propagation effects, and yields Newtonian acceleration through its gradient. In that sense, the existing static development already supplies a definite hierarchy: local displacement, cumulative displacement, and the associated acceleration structure.

What remains to be developed is the dynamical law. At present, this framework does not

derive from deeper substrate microphysics why sustained localized excitation produces the substrate-loading state that appears at effective level as mass sourcing displacement. That sourcing relation therefore remains a foundational postulate. The task of the present paper is to formulate the field equation from which the static equilibrium form $D(r, d)$ emerges as the stationary solution, and to develop the consequences of causal displacement for moving and rotating sources.

The paper therefore does not claim to reduce gravity all the way to a deeper microscopic explanation of why sustained excitation loading displaces the transport structure. Its more limited aim is to formulate the displacement problem at the correct layer and then develop the consequences of that postulate in a causal setting. Once displacement is taken to be physical, and once changes in it are limited by the same propagation speed c that governs the underlying medium, retardation is no longer an added correction. It belongs to the basic structure from the outset. The causal evolution of displacement is therefore the natural setting for Mercury-scale retardation effects, rotating-source response, and the frame-drag branch.

2. Static Equilibrium Geometry and the Cumulative Field

For a stationary isolated source, the displacement problem reduces to an equilibrium geometry. The foundational geometric postulate is that localized source loading inserts volume into the transport structure. In the spherically symmetric case, this means that a source with displacement scale d inserts a volume

$$\frac{4}{3}\pi d^3$$

at the origin of the surrounding transport-supporting medium. The resulting static geometry is then not chosen arbitrarily. It is fixed by the requirement that each spherical shell originally at radius r be displaced outward so as to accommodate the inserted volume while preserving spherical ordering. In the construction adopted here, that displaced shell radius is

$$r' = \sqrt{r^2 + d^2},$$

so the equilibrium displacement factor is the ratio

$$D(r,d) = \frac{r'}{r} = \frac{\sqrt{r^2+d^2}}{r}.$$

Thus the physical geometric postulate is volumetric insertion, while the form of $D(r,d)$ is a geometric consequence. There is no additional functional freedom beyond the source scale d.

This construction makes the local deformation of the transport structure explicit. A unit cell centered at radius r is compressed in the radial direction by a factor $1/D$, stretched in each tangential direction by a factor D, and therefore occupies a net coordinate volume larger by the factor

$$\left(\frac{1}{D}\right) DD = D.$$

The static displacement geometry is therefore anisotropic in its local deformation but scalar in its bookkeeping: the same single function $D(r,d)$ determines radial compression, tangential stretching, and net volume expansion of the displaced medium.

Several immediate features of $D(r,d)$ are worth recording. As $r \to \infty$, one has $D \to 1$, so the geometry tends smoothly to the undeformed background. Near the source, D grows without bound in the unregularized form, reflecting the concentration of the inserted displacement into the innermost region. For the purposes of the present paper, however, the main role of D is the weak-field structure at radii large compared with d. In that regime,

$$D(r,d) = \sqrt{1+\frac{d^2}{r^2}} \approx 1 + \frac{d^2}{2r^2},$$

so the local displacement above background falls with the inverse square of radius.

That inverse-square falloff is the correct local field-strength behavior, but not yet the quantity that governs time dilation and the effective weak-field potential. The key step in this framework is that observable weak-field timing effects are controlled by the **cumulative displacement**, not merely by the local deviation $D-1$. The cumulative field is defined by

$$\Sigma(r) = \int_r^\infty \big(D(r')-1\big)\, dr'.$$

Its physical interpretation is the total excess transport medium encountered along the radial path from r out to infinity. In the weak-field limit $d \ll r$, substituting $D(r)-1 \approx d^2/(2r^2)$ gives

$$\Sigma(r) \approx \int_r^\infty \frac{d^2}{2r'^2}\, dr' = \frac{d^2}{2r}.$$

This is the crucial transition from a local $1/r^2$ structure to an accumulated $1/r$ structure. The framework therefore has a definite hierarchy:

$$D(r) - 1 \sim \frac{d^2}{2r^2}, \qquad \Sigma(r) \sim \frac{d^2}{2r}, \qquad \frac{d\Sigma}{dr} \sim \frac{d^2}{2r^2}.$$

The local displacement gives the field-strength scaling, the cumulative displacement gives the clock and propagation scaling, and the gradient of the cumulative displacement returns the inverse-square acceleration structure.

This hierarchy connects the static geometry to Newtonian gravity. Matching the cumulative displacement to the standard weak-field time-dilation form gives

$$\frac{d^2}{2r} = \frac{GM}{c^2 r},$$

so that

$$d^2 = \frac{2GM}{c^2}.$$

With this identification, the acceleration associated with the cumulative displacement becomes

$$g = c^2 \frac{d\Sigma}{dr} = c^2 \frac{d^2}{2r^2} = \frac{GM}{r^2}.$$

Thus Newtonian gravity appears not as a separate postulate, but as the acceleration associated with the gradient of accumulated displacement once the static source scale is tied to mass through $d^2 = 2GM/c^2$.

3. The Open Dynamics Problem

The static displacement geometry developed in the previous section is already constrained enough to recover the weak-field hierarchy, the cumulative displacement $\Sigma(r)$, and the Newtonian limit. In that sense, the framework is not missing a geometric object. What remains open is the derivation of the dynamical law from which that geometry follows. The function $D(r, d)$ describes the equilibrium displacement around a stationary source, but it is not yet

recovered as the stationary solution of a substrate-derived field equation, action principle, or evolution law.

That open problem should be stated narrowly. The goal is not to replace the full microphysics of matter, nor to absorb all ordinary interactions into a single gravitational master equation. One layer contains the ordinary dynamics of matter and interactions within the transport structure. The other contains the displacement of the transport structure itself. These layers are coupled, but they need not be governed by the same law. The displacement equation belongs specifically to the displacement layer.

Once stated that way, the remaining derivation breaks into three linked but narrower tasks. First, one must explain why sustained localized excitation loads the substrate in a way that sources displacement at all, rather than treating mass-sourced displacement as a separate primitive. Second, one must derive the nonlinear medium-response law for that displacement, so that the present weak-field retarded scalar equation appears as its first-order limit rather than as the full story. Third, one must show how the rotating-source vector branch emerges from the same constitutive structure, so that scalar and vector sectors are not separate constructions patched together after the fact.

The proposal developed later in this paper is that the right intermediate variable is not bare mass itself but a local substrate-loading or occupancy burden, denoted Q_{occ}. In that language, the remaining closure problem is to derive how sustained excitation closure produces Q_{occ}, how Q_{occ} sources cumulative scalar displacement, how the corresponding loading current sources the vector branch, and how the established static geometry $D(r, d)$ is recovered as the stationary solution of that common law. That is a much narrower target than “derive all of gravity from first principles,” and it is the correct one for the present paper.

Gravity is not being treated here as a force internal to the transport structure. It is the transport structure itself being displaced. The task is therefore to propagate information about that displacement causally through the medium, not to reformulate all non-gravitational physics. Sections 4 and 5 develop the first part of that program through the retarded displacement map and its local wave-equation equivalent; Section 8 and Appendix B then identify the constitutive and nonlinear steps that remain genuinely open.

4. Retarded Displacement Map

Once displacement is treated as a physical modification of the transport-supporting network, and once changes in that displacement are taken to propagate at speed c, the natural dynamical construction is retarded rather than instantaneous. The field at a point cannot depend on the source as it is “now” in a coordinate bookkeeping sense, but only on the source configuration whose influence has had time to propagate through the medium. In that sense, the displacement problem is causal from the outset: the geometry at a field point is determined by the past state of the source, evaluated on the relevant propagation cone.

This suggests that the most fundamental statement of the dynamics is not differential but integral. Given a collection of masses with known trajectories, the displacement at each field point is constructed by summing source contributions evaluated at retarded time. In plain terms: given where masses are and how they are moving, evaluate the displacement at each field point by summing retarded contributions from all sources, each propagating at c, using the source position at the retarded time. The importance of this statement is that it defines the dynamical problem directly in physical terms. The field is whatever results from propagation-limited sourcing through the medium; a PDE, if introduced later, is only a local representation of that more basic causal rule.

For a single source, this means the displacement field at spacetime point $(\mathbf{x}, t)$ is determined by the source position $\mathbf{x}_s(t_r)$ at retarded time t_r, where t_r satisfies

$$t - t_r = \frac{R(t_r)}{c}, \qquad R(t_r) = \|\mathbf{x} - \mathbf{x}_s(t_r)\|.$$

The field therefore depends on the delayed source-to-field separation rather than on the instantaneous separation. In the static limit, where $\mathbf{x}_s$ is constant, the retarded construction collapses trivially to the stationary geometry of Section 2. The static displacement function $D(r, d)$ is therefore the equilibrium rest solution of the retarded map, not an unrelated object that later has dynamics added to it.

For many sources, the full field is obtained by combining retarded contributions according to the multi-source composition rule of the framework. In the weak and well-separated regime this may be approximated by a direct sum over retarded source terms. In the fuller displacement picture, however, the contributions are not strictly additive: local space may already be displaced by distant sources before the nearer source acts, so ordering remains relevant. One

is not superposing independent forces in empty space, but composing deformations of a medium already modified by other sources.

This causal picture explains why several previously separate weak-field effects can be treated within a common dynamical language. In the static case, the cumulative displacement $\Sigma(r)$ governs clock-rate and propagation effects, while the local geometry $D(r, d)$ governs spatial deformation. Once the source is time-dependent, those same quantities become retarded fields. Mercury precession can then be read as a consequence of finite-speed response on an accelerated orbit; Shapiro delay remains the integrated clock-rate effect of the cumulative displacement along the signal path; and the rotating-source problem becomes a retarded field problem rather than a purely static geometric one.

For that reason, the retarded map should be treated here as the conceptual heart of the dynamics paper. It states what gravity is at the displacement layer: not an instantaneous action, and not a force law internal to matter dynamics, but a propagation-limited modification of the transport stage sourced by the past state of matter.

5. Two-Layer Coupled Structure and the Coordinate-Level Effective Geometry

The retarded displacement map gives the physical rule: mass-sourced displacement propagates through the transport-supporting medium at speed c, and the field at any point is determined by the retarded source configuration rather than by instantaneous source position. The present section develops two further steps. First, the retarded map is given an explicit integral form whose static limit recovers the established cumulative displacement. Second, the displacement is shown to define an effective coordinate-level geometry whose weak-field orbit structure reproduces the standard perihelion precession.

5.1 Two coupled layers

The dynamics of this framework are organized as two coupled layers rather than a single master equation.

The first layer contains the ordinary dynamics of excitations — matter as sustained LC oscillations, electromagnetic fields, and their interactions — evolving within the transport-supporting structure. These excitations propagate through the transport structure at the speed

set by the vacuum storage medium: $c = 1/\sqrt{\varepsilon_0 \mu_0}$. Their energy-momentum distribution is the source of displacement.

The second layer contains the displacement of the transport structure itself. The cumulative displacement field $\Sigma(\mathbf{x}, t)$ responds to the energy distribution of the first-layer excitations, with changes propagating causally at c. The resulting displacement modifies the effective geometry through which first-layer excitations propagate. Gravity is therefore not a force internal to the excitation layer. It is a modification of the stage on which those excitations evolve, fed back from the displacement layer.

The coupling is bidirectional. Excitation energy sources displacement; displacement reshapes the transport geometry that governs excitation dynamics. In the general case, the two layers must be solved together: moving sources change the displacement, and the changing displacement modifies the source trajectories.

In the static and weak-field regimes, however, this coupling decouples into a one-way hierarchy. When the source is stationary, the displacement layer reduces to a Poisson equation that is solved once for the given energy distribution. The resulting $\Sigma(r)$ sets a fixed background geometry, and the excitation layer then evolves on that background without modifying it. The two layers do not iterate. This static decoupling is what makes the equilibrium geometry tractable: the source sits at rest, its energy distribution sources $\Sigma(r) = d^2/(2r)$, and the resulting displacement geometry is the equilibrium form $D(r, d)$ already established.

The same decoupling extends to test excitations on a static background. A planet orbiting the Sun moves within the Sun's fixed displacement geometry. The planet's own displacement is negligible compared to the Sun's, so it does not modify the background. The displacement layer is solved once for the dominant source; the excitation layer provides the orbit equation on that fixed background. All of the weak-field results developed in this paper — Mercury precession, Shapiro delay, light deflection, and the GPB geodetic and frame-drag measurements — belong to this decoupled regime.

The layers re-couple when comparable-mass sources move and accelerate in each other's displacement fields. A compact binary system, where each body's displacement significantly affects the other's trajectory and the trajectories in turn modify the displacement, requires the full retarded integral with both layers evolving together. That strong-coupling regime is the frontier beyond the present paper.

5.2 The displacement layer: retarded integral form

The natural variable for the displacement layer is the cumulative displacement $\Sigma(\mathbf{x}, t)$ rather than the local displacement factor D. The reason is the same one already seen in the static theory: $D - 1$ carries the local $1/r^2$ field-strength structure, while Σ carries the accumulated $1/r$ structure that governs clock-rate, propagation, and weak-field dynamics.

The retarded map of Section 4, written as an explicit integral, gives

$$\Sigma(\mathbf{x}, t) = \frac{G}{c^2} \int \frac{\rho_{\mathrm{LC}}(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|} \, d^3x',$$

where ρ_{LC} is the energy density of the LC excitations that constitute the source, and the retarded time satisfies $t_r = t - |\mathbf{x} - \mathbf{x}'|/c$. This is the displacement analog of the retarded potential in electromagnetism: the cumulative displacement at a field point is sourced by the retarded energy distribution, propagated through the vacuum medium at speed c.

The integral form is the more fundamental statement. Its equivalent local differential representation is

$$\frac{1}{c^2}\partial_t^2 \Sigma - \nabla^2 \Sigma = \frac{4\pi G}{c^2} \, \rho_{\mathrm{LC}},$$

which is a wave equation with source, propagating at the medium speed c. The propagation speed is not a free parameter: it is the only speed the vacuum storage medium supports. In the static limit, this reduces to

$$\nabla^2 \Sigma = -\frac{4\pi G}{c^2} \, \rho_{\mathrm{LC}},$$

whose point-source solution is $\Sigma(r) = GM/(c^2 r) = d^2/(2r)$, recovering the established cumulative displacement. Once $\Sigma(\mathbf{x}, t)$ is known, the local geometric displacement factor may be recovered algebraically by

$$D(r, t) = \frac{\sqrt{r^2 + 2r\,\Sigma(r, t)}}{r}.$$

For the static point-source solution, this reduces immediately to the equilibrium form of Section 2.

The displacement field does not have independent oscillatory dynamics in the way that a free

wave does. It tracks the retarded source distribution. Outgoing displacement disturbances — the framework's gravitational waves — are the retarded response to accelerating matter, not free oscillations of the displacement field itself. The wave-equation form encodes the same causal retardation as the integral; the two descriptions are complementary.

5.3 The excitation layer: effective coordinate geometry

The second step is to determine what geometry the first-layer excitations see when propagating through a displacement background. The key constraint comes from the vacuum storage structure. The storage capacity $F \cdot H = \varepsilon_0 \mu_0$ is attached to the coordinate grid (Appendix A). Displacement moves the transport structure relative to that grid, but it does not move the grid itself.

In the displaced transport structure, a unit cell at radius r is compressed radially by $1/D$ and stretched tangentially by D. A physical meter — realized within transport as a sustained LC excitation — co-deforms with the transport. The physical meter is not identical to the transport structure; it is an observable realized within it. When transport is displaced, the rod's realized properties change accordingly: radially, the physical meter spans fewer coordinate meters; tangentially, it spans more.

The question is how this deformation appears in the coordinate-level description that governs the effective dynamics. The answer is set by which quantities belong to coordinates and which belong to transport.

Temporal component. The 3D oscillation that constitutes a clock cycle traverses a net volume of $F \cdot H$ that scales as D relative to the undisplaced case (the product of radial compression $1/D$ and tangential stretch D in two directions). A complete cycle therefore encounters more total storage, and the clock runs slower. The realized clock rate relative to coordinate time is $1/(1+\Sigma)$, giving an effective time-time component

$$f(r) = \frac{1}{(1+\Sigma(r))^2}.$$

Radial component. A physical meter in the radial direction spans $1/D$ coordinate meters. Equivalently, one coordinate meter radially spans D physical meters of transport. In the effective coordinate-level line element, this means the radial coefficient is

$$g(r) = D(r)^2 = 1 + 2\Sigma(r),$$

where the second equality uses the weak-field relation $D^2 \approx 1 + 2\Sigma$.

Angular component. The tangential stretching of transport cells means each cell covers more coordinate arc. But the coordinate arc $r\,d\theta$ is a property of the coordinate grid, and the coordinate grid is not displaced. The angular bookkeeping in coordinates remains Euclidean. This is the critical point: the storage capacity $F \cdot H$ per coordinate meter is unchanged by displacement, and the coordinate angles are defined on the coordinate grid, not on the transport chart. The angular coefficient is therefore

$$h(r) = 1.$$

The effective coordinate-level line element for excitations propagating through the static displacement background is then

$$ds^2 = -\frac{c^2}{(1+\Sigma)^2}\,dt^2 + D^2\,dr^2 + r^2\,d\Omega^2.$$

In the weak field, with $\sigma = r_s/r = 2GM/(c^2 r) = 2\Sigma$, this becomes

$$ds^2 \approx -(1-\sigma)\,c^2\,dt^2 + (1+\sigma)\,dr^2 + r^2\,d\Omega^2,$$

which is the standard weak-field form to leading order in σ.

5.4 Recovery of the weak-field orbit structure

The effective geometry of the preceding subsection is not merely a rewriting of known results. It is derived from the displacement framework's own internal logic: the storage capacity is on the coordinate grid, the displacement deforms transport relative to that grid, and the physical meter co-deforms with transport. The fact that the resulting line element agrees with the standard weak-field Schwarzschild form at leading order is a consistency check, not a circular input.

The orbit equation for a test excitation on this background may be derived by standard methods. In the weak-field expansion, the orbit equation takes the form

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2}\,u^2 + O(\sigma^2),$$

where $u = 1/r$ and L is the conserved angular momentum per unit mass. The $3GM\,u^2/c^2$ correction is the term responsible for perihelion precession. For Mercury, it gives

$$\delta\omega = \frac{6\pi GM}{a(1-e^2)\,c^2} = 42.98 \text{ arcsec/century.}$$

The three contributions previously identified for Mercury precession — retardation mismatch, external-field kinematic steering, and cumulative displacement gradient — are therefore not three separate physical mechanisms to be added independently. They are three aspects of a single geometric fact: the displacement of transport on the coordinate grid modifies the time component and the radial component of the effective coordinate geometry, while the angular component remains Euclidean. The orbit equation on that effective geometry produces the correct precession as one integrated result rather than as a sum of separately motivated corrections. The interpretive decomposition is $\alpha_1 = 1$ (retardation), $\alpha_2 = 1/2$ (external-field kinematic steering at the standard Thomas coefficient), $\alpha_3 = 3/2$ (displacement geometry), totaling $\alpha = 3$.

The same effective geometry also supports the Shapiro delay as a pure clock-rate effect and the geodetic precession applied to the GPB orbit [7]. In each case, the coordinate-level description carries the full physics. Local clock rates, rod lengths, and other realized-geometry observables may be recovered from the coordinate description whenever needed, but the dynamics are most cleanly posed in coordinate terms.

6. Rotating Sources and the Vector Response

The preceding sections establish the scalar displacement sector: at the effective level, mass-energy-sourced displacement is causal, propagation-limited, and in the stationary spherically symmetric limit is captured by the equilibrium geometry $D(r, d)$ or, more naturally for dynamics, by the cumulative field $\Sigma(\mathbf{x}, t)$. That scalar sector is sufficient for the static point-source problem and for the first weak-field timing and propagation effects. It is not sufficient for the first genuinely non-spherical source class: rotating bodies. The rotating-source problem requires the causal displacement framework to develop a corresponding vector response.

This extension follows naturally from the retarded-map viewpoint. In the static scalar case, the source is fully specified by radial displacement around a stationary center. For a rotating body, however, the source elements are not merely displaced in radius; they are moving around the source axis with definite handedness. The retarded field at a probe point must therefore retain information about source motion as well as source location. A purely scalar field can encode the strength of cumulative displacement, but it cannot by itself encode the azimuthal

asymmetry associated with source rotation. The causal displacement layer must therefore admit a vector channel in addition to the scalar cumulative field.

The relevant object is the retarded **vector** field built from the spin-odd azimuthal component of the source pull. This is the key conceptual step: the rotating-source problem belongs to the vector response of the displacement field, not to the symmetric scalar geometry alone. The leading far-field form is taken to be

$$V_{\phi,\mathrm{odd}}(\rho,\theta) \propto \omega \, \frac{\cos\theta}{\rho^2},$$

where ω is the source spin, ρ is the probe radius, and θ is latitude relative to the spin axis. This form is linear in source spin, odd under reversal of ω, azimuthal rather than radial, and modulated by latitude through $\cos\theta$.

This vector channel should be introduced carefully. The paper does not need to claim that the full rotating-source field equation has already been derived in closed substrate form. What can be said more cleanly is that the retarded displacement map, once applied to a spinning source, naturally generates a vector response whose stable observational content lies in the spin-odd azimuthal sector. The scalar cumulative field remains the correct object for the stationary weak-field timing structure, while the rotating-source response requires the corresponding vector extension of the same causal architecture.

7. Gravity Probe B and the Frame-Drag Estimate

The rotating-source vector response becomes physically meaningful only when it is tied to an observable. In this framework, the natural first test case is Gravity Probe B. The role of the present section is therefore not to introduce a new phenomenological domain, but to extract the first concrete weak-field observable from the rotating-source displacement field developed in the previous section.

The observable to be extracted is the frame-drag component of the GPB gyroscope drift. The geodetic branch follows from the effective coordinate geometry as a single integrated result and need not be rederived in the body. The present section is concerned only with the rotating-source contribution.

Within the rotating-source construction, the relevant field is the retarded spin-odd azimuthal

component

$$V_{\phi,\mathrm{odd}}(\rho,\theta) \propto \omega\,\frac{\cos\theta}{\rho^2}.$$

This object is the stable carrier of the frame-drag signal in the current route. Its symmetry already fixes much of the observable structure. The response is linear in the source spin ω, reverses sign under spin reversal, and varies over latitude through the factor $\cos\theta$. The azimuthal character of the field is what makes it the natural home of frame dragging in this framework: it is a genuine rotating-source response, not a correction to the scalar equilibrium geometry.

The practical extraction proceeds in four steps. First, determine the equatorial fixed-probe value at the GPB orbital radius. Second, extend that value over latitude using the established $\cos\theta$ dependence. Third, perform the polar-orbit projection analytically rather than by stepping a moving probe around the full orbit. Fourth, convert the thin-shell source model to the Earth's actual moment of inertia.

On that route, the thin-shell estimate for the GPB frame-drag signal is about 82.5 milliarcsec/yr. Applying the Earth moment-of-inertia correction

$$\frac{I_\oplus/(MR^2)}{2/3} = \frac{0.3307}{2/3} \approx 0.496,$$

yields an Earth-corrected prediction of about 40.9 milliarcsec/yr. This is close to the GPB measured frame-drag value of 39.2 milliarcsec/yr. In the logic of the present paper, that agreement is not yet presented as a final closed derivation from the deeper substrate, but it is strong evidence that the rotating-source displacement field is carrying the right weak-field structure.

8. Limits, Status, and the Remaining Closure Problem

The preceding sections establish a definite intermediate result. This framework already supplies a static equilibrium geometry for mass-sourced displacement, a cumulative field Σ with the required $1/r$ weak-field structure, and a causal retarded interpretation in which displacement changes propagate at the medium speed c. The two-layer coupled structure then

provides the dynamical organization: the displacement layer tracks the retarded source distribution through a wave equation at speed c, while the excitation layer evolves within the effective coordinate geometry set by that displacement. The coordinate-level line element — with its time dilation from Σ, radial modification from D^2, and unmodified angular sector — recovers the correct weak-field orbit structure, including the standard perihelion precession. The rotating-source analysis identifies a spin-odd azimuthal vector response as the natural home of frame dragging.

At the same time, the limits of the present result should be stated plainly. The foundational effective-level postulate remains exactly that: localized source loading inserts volume into the transport structure, and changes in that displacement propagate causally. What has not yet been derived is the deeper microphysical reason that sustained excitation closure has this sourcing role. The framework therefore remains explicit about where it is postulating and where it is deriving. The geometry of $D(r, d)$, the cumulative displacement $\Sigma(r)$, and the weak-field hierarchy are derived once the postulate is granted; the postulate itself is not yet reduced to a deeper mechanism.

The two-layer structure and the retarded integral for $\Sigma(\mathbf{x}, t)$ represent a step beyond the purely static geometric development. The retarded map is the more fundamental statement: source information propagates outward at c, and the field at a point depends on the retarded source configuration. The wave equation is its local differential equivalent. Together they fix the propagation speed to c (the only speed the medium supports) and, at the effective level, tie the source term directly to the LC energy density of matter. At the deeper constitutive level introduced later, the corresponding substrate-loading source is denoted by Q_{occ}. The effective coordinate geometry derived from the displacement then recovers the weak-field tests without requiring separately motivated correction terms.

The rotating-source case sharpens this point. The scalar cumulative field is sufficient for the stationary weak-field timing sector, but it is not sufficient for rotating bodies. There the framework requires a vector response, and the current route identifies the stable observable channel as the spin-odd azimuthal component of the retarded field. That is enough to support a coherent GPB route, but it is not yet the same thing as a fully derived rotating-source field equation. The present paper therefore claims the vector channel as the correct qualitative and quantitative direction, not as a fully closed final theory of spinning sources.

The remaining closure step can now be stated more precisely than before. A completed displacement dynamics would need to do at least four things. First, it would have to derive the

displacement sourcing relation from deeper substrate structure rather than leave it primitive. Second, it would have to supply a full retarded field law or equivalent local equation whose stationary solution is the established $D(r, d)$ geometry. Third, it would have to incorporate the vector response of moving and rotating sources in the same unified formalism rather than as a separate downstream construction. Fourth, it would have to recover the already-established weak-field tests without changing their successful static or weak-field sectors.

Beyond the weak-field orbital tests, the two-layer structure also implies gravitational radiation from accelerating sources. The displacement deformation is not merely a scalar but carries a tensor structure reflecting the anisotropy of the transport deformation (radial compression, tangential stretch). For a compact binary, the time-varying tensorial pattern radiates at the quadrupole level. The scalar wave equation for the cumulative displacement Σ gives one sixth of the general-relativity power; the full symmetric deformation tensor has six independent components, each radiating with equal power due to the isotropy of the vacuum impedance; and the total power therefore matches the GR quadrupole formula exactly [7]. For circular orbits this reproduces the Hulse-Taylor inspiral rate. For eccentric orbits, the oscillating trace provides an additional breathing-mode channel absent in general relativity, potentially modifying the eccentricity enhancement function $f(e)$.

8.1 Self-consistency of the two-layer loop

The two-layer structure raises a natural self-consistency question. The displacement layer sources Σ from the energy density of LC excitations. The excitation layer evolves within the effective geometry set by that displacement. For the loop to close, the energy distribution of a source at rest in its own displacement field must be consistent with the Σ it generates.

In the static case, this loop closes straightforwardly. A point mass with energy mc^2 sources a cumulative displacement $\Sigma(r) = GM/(c^2 r)$ through the Poisson equation. The resulting effective geometry is the coordinate-level line element of Section 5.3. A stationary source sitting at the origin of that geometry is at rest in the displacement it creates — there is no further feedback at this order. The static displacement is the equilibrium configuration, and the effective geometry is self-consistent.

The remainder of this subsection and the next sketch how the self-consistency loop might close at a deeper level. These remarks identify where the connections must eventually be made, but the derivations themselves are not completed in this paper.

The deeper question is what determines the source energy mc^2 itself. In this framework, matter is a sustained LC excitation of the vacuum medium. Such an excitation cycles between the capacitive (ε_0) and inductive (μ_0) storage modes at a definite frequency. The resonant frequency of a vacuum region of spatial extent ℓ is $f \sim c/\ell$, set entirely by the medium properties. A sustained excitation — one that persists rather than dispersing — must satisfy a closure condition: after one complete LC cycle, the excitation must return to its starting configuration. This topological closure quantizes the action per cycle, and that quantum is Planck's constant h. The energy of the excitation is then $E = hf$, and its spatial extent is the Compton wavelength $\ell_C = h/(mc)$.

The same LC substrate also determines the de Broglie wavelength of a moving excitation. A particle at rest has a spatially uniform internal phase. When the excitation moves at velocity v through the transport medium, the simultaneity slope v/c^2 per unit length — already derived in the homogeneous propagation framework — converts that uniform phase into a spatial phase gradient. The wavelength of that gradient is

$$\lambda_{\mathrm{dB}} = \frac{c^2}{v\, f_0} = \frac{h}{mv} = \frac{h}{p},$$

where $f_0 = mc^2/h$ is the rest-frame cycling frequency. The de Broglie relation is therefore not an independent postulate. It is the spatial phase structure of an LC resonance viewed through the simultaneity tilt of the propagation framework.

8.2 Hierarchy of constants

The framework's constant structure then has a natural hierarchy. The medium properties ε_0 and μ_0 give the propagation speed c and the vacuum impedance Z_0. The closure condition on sustained LC resonances gives the quantum of action h. The gravitational constant G is the bridge between the excitation layer and the displacement layer: it determines how much displacement one unit of LC energy produces. The relationship $d^2 = 2Gm/c^2$ connects the displacement scale of a source to its mass, and the per-nucleon form $G = d_n^3 c^2/m_n$ (times a geometric factor) suggests that G is derivable if the nucleon's displacement radius d_n can be determined from the LC resonance structure of the nucleon itself.

The self-consistency of the full loop — medium properties determine resonance structure, resonance structure determines source energy, source energy determines displacement, displacement modifies the medium geometry, and the modified geometry supports the

resonance — is not yet closed in this paper. But the two-layer structure developed here defines each link in the chain and identifies where the remaining derivations must connect.

8.3 Provisional constitutive route from excitation loading to displacement

The remaining closure problem may be stated more precisely. The present paper still treats the displacement sourcing relation as primitive: mass-energy sources displacement, and changes in that displacement propagate causally at the medium speed c. What is missing is the deeper substrate reason that a sustained localized excitation should have this sourcing role at all. Once that sourcing rule is granted, the static geometry $D(r, d)$, the cumulative field Σ, and the weak-field observational hierarchy are already fixed. The unresolved step is the constitutive bridge from excitation structure to displacement source.

A natural intermediate move is to introduce an explicit substrate-loading variable. Let $Q_{\mathrm{occ}}(\mathbf{x}, t)$ denote the local occupancy burden placed on the vacuum storage structure by a sustained LC excitation. It is a dimensionless measure of how strongly the local capacitive-inductive support is engaged by closure. In the unloaded background, $Q_{\mathrm{occ}} = 0$. In the presence of a sustained localized excitation, $Q_{\mathrm{occ}} > 0$. The physical claim is then that displacement is not sourced by bare mass as an unexplained primitive, but by the cumulative retarded response of the transport structure to this local occupancy burden.

This proposal is consistent with the broader architecture already in place. The vacuum medium supplies the propagation speed through its storage densities ε_0 and μ_0. The excitation layer supplies sustained localized LC cycling. The missing step is to show that sustained closure also occupies or burdens the same storage structure in a way that alters the transport support seen by other excitations. In that language, the source relation becomes a constitutive statement: localized excitation closure loads the substrate, and the displacement field is the retarded response to that loading.

The weak-field retarded map already has the form

$$\Sigma(\mathbf{x}, t) = \frac{G}{c^2} \int \frac{\rho_{\mathrm{LC}}(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|} \, d^3x'.$$

The present subsection interprets that relation as the first-order limit of a more basic constitutive law,

$$Q_{\rm occ} = Q_{\rm occ}(\rho_{\rm LC}, \mathcal{C}, \mathcal{E})\,,$$

where $\mathcal{C}$ denotes closure state and $\mathcal{E}$ denotes environmental state. In the weak field, where the excitation is only a small perturbation of the background and environmental feedback is negligible, the leading-order relation should reduce to

$$Q_{\rm occ} \approx \kappa\, \rho_{\rm LC}.$$

The cumulative displacement may then be written as the retarded response to occupancy burden,

$$\Sigma(\mathbf{x}, t) = \Lambda \int \frac{Q_{\rm occ}(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|}\, d^3x',$$

with $\Lambda\kappa = G/c^2$, immediately recovering the established weak-field source law. On this reading, the statement “mass-energy sources displacement” is not abandoned, but sharpened: sustained excitation closure loads the substrate locally, and the cumulative retarded response of that loading is the displacement field.

This constitutive reinterpretation also clarifies the nonlinear strong-field problem. Section B.4 already argues that the full displacement equation must acquire state-dependent coefficients because an already displaced medium responds differently to further displacement. If the more basic state variable is $Q_{\rm occ}$, then the cleaner nonlinear form is not merely a wave equation with coefficients depending on Σ, but a medium equation whose coefficients depend on the underlying occupancy burden that is doing the loading:

$$\frac{1}{c_{\rm eff}(Q_{\rm occ})^2}\, \partial_t^2 \Sigma - \nabla\cdot\big[\alpha(Q_{\rm occ})\, \nabla\Sigma\big] = S[Q_{\rm occ}].$$

In the weak field, $Q_{\rm occ} \ll 1$, $c_{\rm eff} \to c$, $\alpha \to 1$, and the source reduces to the linear equation of Section 5.2. In the strong field, the constitutive burden modifies the effective stiffness of the medium and therefore the propagation law itself. This does not yet determine the explicit forms of $c_{\rm eff}$, α, or $S[Q_{\rm occ}]$, but it sharpens what remains to be derived: not an arbitrary nonlinear correction, but the state dependence implied by substrate loading.

The same constitutive move also suggests the natural origin of the vector sector. A static source is characterized by occupancy density $Q_{\rm occ}$. A moving or rotating source carries an occupancy current,

$$\mathbf{J}_{\mathrm{occ}} = Q_{\mathrm{occ}}\,\mathbf{v}.$$

The scalar displacement field is then the retarded response to occupancy density, while the vector displacement field is the retarded response to occupancy current. In that form, the rotating-source branch is no longer a separate add-on introduced after the scalar field has already been written down. Both sectors arise from one constitutive source structure: occupancy burden and occupancy flow.

This subsection does not claim that the closure problem is solved. The closure functional that permits sustained localization remains open. The constitutive map $Q_{\mathrm{occ}}(\rho_{\mathrm{LC}}, \mathcal{C}, \mathcal{E})$ is not yet known in explicit form. The nonlinear coefficients of the medium response are not yet derived. But the remaining derivation target is now narrower and more definite. A completed theory would need to derive the closure burden of a sustained excitation, derive the corresponding occupancy variable Q_{occ}, derive the scalar and vector retarded displacement laws as the medium response to Q_{occ} and $\mathbf{J}_{\mathrm{occ}}$, and recover the established equilibrium geometry $D(r, d)$ as the stationary solution of that common law.

8.4 Local scalar--vector field system for occupancy-sourced displacement

The preceding sections motivate a more definite intermediate closure target for the displacement layer. The remaining problem is no longer to guess whether a unified law should exist, but to identify the local source variables and the corresponding field equations from which both the static scalar sector and the rotating vector sector arise. In the present framework, the natural constitutive source is the local substrate-loading variable Q_{occ}, interpreted as the fraction of local vacuum storage support engaged by sustained excitation closure. The scalar displacement response is then sourced by occupancy density, while the rotating branch is sourced by occupancy current. This preserves the established weak-field structure while making the deeper source architecture more explicit.

Let

$$Q_{\mathrm{occ}}(\mathbf{x}, t)$$

denote the local closure burden imposed on the substrate by sustained excitation. In the weak-loading regime, the simplest constitutive identification is

$$Q_{\mathrm{occ}} \approx \kappa\, \rho_{\mathrm{LC}},$$

so that the present energy-density sourcing of the displacement layer is recovered as the first-order limit of a deeper occupancy description. The scalar cumulative field Σ is then governed locally by

$$\frac{1}{c^2}\, \partial_t^2 \Sigma - \nabla^2 \Sigma = \lambda_s\, Q_{\mathrm{occ}},$$

with retarded solution

$$\Sigma(\mathbf{x}, t) = \frac{\lambda_s}{4\pi} \int \frac{Q_{\mathrm{occ}}(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|}\, d^3 x'.$$

Matching this to the established weak-field displacement law,

$$\frac{1}{c^2} \partial_t^2 \Sigma - \nabla^2 \Sigma = \frac{4\pi G}{c^2} \rho_{\mathrm{LC}},$$

fixes the weak-limit normalization

$$\lambda_s\, \kappa = \frac{4\pi G}{c^2}.$$

The familiar retarded integral for Σ is therefore recovered immediately once Q_{occ} is taken to be proportional to ρ_{LC} in the dilute-loading limit.

The rotating sector follows from the same source logic. Once the source carries occupancy density, a moving or rotating source necessarily carries occupancy current. Define

$$\mathbf{J}_{\mathrm{occ}}(\mathbf{x}, t) = Q_{\mathrm{occ}}(\mathbf{x}, t)\, \mathbf{v}(\mathbf{x}, t).$$

The vector displacement field $\mathbf{A}$ is then governed, in the same weak-limit approximation, by

$$\frac{1}{c^2}\, \partial_t^2 \mathbf{A} - \nabla^2 \mathbf{A} = \lambda_v\, \mathbf{J}_{\mathrm{occ}},$$

with retarded solution

$$\mathbf{A}(\mathbf{x},t) = \frac{\lambda_v}{4\pi} \int \frac{\mathbf{J}_{\mathrm{occ}}(\mathbf{x}', t_r)}{|\mathbf{x}-\mathbf{x}'|}\, d^3x'.$$

Using $Q_{\mathrm{occ}} \approx \kappa \rho_{\mathrm{LC}}$ gives

$$\mathbf{J}_{\mathrm{occ}} \approx \kappa\, \rho_{\mathrm{LC}} \mathbf{v},$$

so matching to the existing retarded vector construction fixes

$$\lambda_v\, \kappa = \frac{4\pi G}{c^3}.$$

In this form, the vector branch is no longer a separate add-on introduced only for rotating bodies; it is the current-sector completion of the same constitutive source law that generates the scalar displacement field.

The pair $(Q_{\mathrm{occ}}, \mathbf{J}_{\mathrm{occ}})$ also suggests the natural source-balance equation

$$\partial_t Q_{\mathrm{occ}} + \nabla \cdot \mathbf{J}_{\mathrm{occ}} = \Gamma_{\mathrm{occ}},$$

where Γ_{occ} represents local creation, sustainment, or relaxation of closure burden. This need not be an exact conservation law in the same sense as charge conservation, because substrate occupancy is not yet known to be a strictly conserved quantity. It is, however, the correct structural equation for relating stationary, translating, rotating, and time-varying sources within one common source description. In the static limit, $\mathbf{J}_{\mathrm{occ}} = 0$ and $\partial_t Q_{\mathrm{occ}} = 0$, so the scalar field reduces to the stationary occupancy-sourced displacement field. For moving sources, the current branch becomes nonzero and supplies the missing handedness and azimuthal structure required for frame dragging.

At the metric level, this scalar--vector split maps naturally onto the effective geometry already used elsewhere in the paper. The scalar field Σ governs the isotropic timing and radial deformation sector, while the vector field contributes the off-diagonal transport-twist terms. In a schematic generalized local scalar--vector embedding,

$$ds^2 = -\frac{c^2}{(1+\Sigma)^2} dt^2 + D^2\, \delta_{ij}\, dx^i dx^j + 2\, \beta_i\, dt\, dx^i,$$

with
[
\beta_i \equiv \chi_A A_i,

with χ_A a proportionality constant fixing the metric image of the vector field.
]
For slow axisymmetric rotation, only the azimuthal component survives, so that the generalized shift-like term reduces to the familiar $g_{t\phi}$ structure already developed in the rotating appendix. In this sense, the existing slow-rotation metric is the weak-loading geometric image of the occupancy-current sector. Earlier spin-odd azimuthal observables such as $V_{\phi,\mathrm{odd}}$ should therefore be read as extracted components of this same vector sector rather than as independent objects.

This local scalar--vector system does not yet close the framework. Three elements remain to be derived from the substrate itself: the constitutive law $Q_{\mathrm{occ}}(\rho_{\mathrm{LC}}, \mathcal{C}, \mathcal{E})$, the source-balance term Γ_{occ}, and the nonlinear loaded-medium response functions that replace the weak-limit coefficients in the strong field. Nevertheless, the structure above narrows the remaining derivation problem substantially. The source architecture is no longer merely ``mass sources scalar displacement, and rotation adds a vector term." It is instead: sustained excitation closure imposes substrate occupancy; occupancy density sources the scalar displacement field; occupancy current sources the vector displacement field; and both sectors are to be recovered as limits of one common loaded-medium law.

9. Conclusion

This framework treats gravity as a causal displacement of the transport-supporting spatial network rather than as a force internal to ordinary matter dynamics. Starting from the effective-level postulate that sustained excitation loading sources that displacement, and that changes in it propagate at speed c, the framework develops a retarded displacement map, an effective coordinate geometry, and a rotating-source response within one common causal picture.

Its present status should be stated precisely. The framework is not limited to weak-field

geometry. The static equilibrium displacement $D(r, d)$ is nonperturbative, and the corresponding effective geometry is defined by the same displacement logic at all field strengths. The rapid-rotation equilibrium construction likewise belongs to that nonperturbative geometric layer. What has mainly been carried through in explicit observational detail is the weak-field regime, because that is where current solar-system and near-Earth tests lie and where the leading-order cumulative field Σ and linearized propagation law are sufficient.

The dynamical problem is organized as two coupled layers. The displacement layer tracks the retarded source structure through the medium at speed c, with the retarded integral as the fundamental statement and the wave equation as its local differential equivalent in the weak regime. The excitation layer contains the ordinary dynamics of matter and fields evolving within the effective coordinate geometry set by the displacement. That geometry is determined by the framework's own internal logic: the vacuum storage capacity $F \cdot H$ is anchored to the coordinate grid, the displacement deforms transport relative to that grid, and the physical meter co-deforms with transport. The resulting coordinate-level line element modifies the time component and the radial component while leaving the angular sector Euclidean, and its weak-field orbit equation recovers the standard perihelion precession as a single integrated geometric result.

This same causal architecture extends beyond the stationary scalar sector. For rotating sources, the relevant response is not exhausted by the symmetric cumulative field alone, but requires a spin-odd azimuthal vector channel. In the current route, that channel yields a Gravity Probe B frame-drag estimate of about 40.9 mas/yr after polar-orbit projection and Earth moment-of-inertia correction, placing the framework on the observed weak-field frame-drag scale.

The result is not yet a final closed field theory, but the remaining gap is now narrower than a generic appeal to “more strong-field work.” The missing step is the constitutive derivation of the loaded-medium law: derive how sustained excitation closure produces the substrate occupancy burden Q_{occ}, derive how that burden modifies the scalar and vector retarded response of the medium, recover the established static geometry $D(r, d)$ as the stationary solution of the full law, and thereby unify the scalar and rotating-source sectors within one formal dynamical framework. In that sense, the framework already contains nonperturbative geometric structure, while the genuinely open problem is the derivation of the nonlinear strong-field dynamics from the deeper substrate.

Appendix A. Vacuum Storage, Propagation Units, and Meters per Storage Capacity

This framework treats the vacuum storage field as a distributed substrate support characterized by the familiar constants ε_0 and μ_0, with units F/m and H/m. Their product sets the propagation scale,

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}},$$

so that $\varepsilon_0 \mu_0$ carries dimensions of time-squared per length-squared. In this sense, the propagation speed is not an additional postulate but the wave speed supported by the storage structure itself.

The important geometric point for the displacement framework is that the storage capacity represented by $F \cdot H$ is attached to the coordinate description. When source loading displaces the transport structure, a unit cell of the transport-supporting medium spans more or less coordinate extent depending on direction, but the local physical meter realized by rods and clocks co-expands or co-compresses with that same displacement. The result is that the amount of storage structure encountered per physical meter remains matched to the realized meter itself. Light therefore continues to propagate at C in the locally realized geometry even in a displaced region; what changes is not the local physical speed of light but the realized clock rate and the accumulated displacement along a path.

This point is used repeatedly in the appendices that follow. It underlies the interpretation of Shapiro delay as a pure clock-rate effect, supports the cumulative-displacement reading of weak-field timing structure, and explains why geometric displacement and timing effects can be separated without introducing a local anisotropy in the speed of light.

Appendix B. Nonlinear Displacement and the Strong-Field Regime

The wave equation for Σ developed in Section 5.2 is linear: the propagation speed is c, the source term is the LC energy density, and the superposition of solutions from multiple sources is additive. This linearity is valid in the weak field, where $\Sigma \ll 1$ and the displacement is a small perturbation of the transport structure. The present appendix identifies where that

linearity breaks down and what replaces it.

B.1 Displacement-dependent stiffness

The linear wave equation treats the transport medium as having fixed properties: the $F \cdot H$ storage per coordinate meter is constant, and the propagation speed c is determined by that constant storage density. But when the transport structure is displaced, the medium is already deformed. A region with existing displacement $\Sigma > 0$ has its transport cells stretched tangentially and compressed radially. Further displacement into that already-deformed medium encounters less resistance than the same displacement would in the undisplaced background.

The physical mechanism is the same one that produces the golden-ratio crossing noted earlier: the displacement function $D(r)$ describes a medium whose local properties depend on the current deformation state. In the language of the wave equation, the coefficient multiplying $\nabla^2\Sigma$ is not the bare c^2 but an effective speed that depends on the local value of Σ. The displacement equation is therefore inherently nonlinear: the medium through which displacement propagates is itself modified by the displacement.

In the weak field this correction is negligible. At the surface of the Sun, $\Sigma \sim 10^{-6}$, and the nonlinear correction to the propagation coefficient is of order $\Sigma^2 \sim 10^{-12}$. At any currently accessible precision level, the linear wave equation suffices. The nonlinearity becomes significant only when Σ approaches order unity — the strong-field regime.

B.2 Multi-source composition and the ordering rule

For multiple sources, the nonlinearity appears more concretely. The displacement from each source acts on a medium that is already displaced by every other source. In the framework's composition rule, contributions are applied in order of distance: the farthest source first, the closest source last. Each successive displacement acts on the result of the previous one, not on the undisplaced background.

The consequence is that two nearby sources allow each other to displace more than each would in isolation. The first source displaces the medium; the second encounters an already-stretched medium with reduced stiffness and therefore produces a slightly larger displacement than it would in flat space. The combined displacement of two nearby sources is slightly greater than the sum of their individual displacements. This is the origin of the second post-Newtonian parameter $\beta \neq 1$ in the framework, though the correction is of order Σ^2 and is

unmeasurably small in any current weak-field test.

B.3 Strong-field structure

The strong-field regime should be separated into two distinct claims. First, the **equilibrium displacement geometry** itself is already specified nonperturbatively by the framework through the exact displacement factor $D(r, d) = \sqrt{r^2 + d^2}/r$ and its corresponding deformation picture. Second, the **time-dependent strong-field dynamics** that would govern how such configurations form, evolve, and interact are not yet derived from the LC substrate. The discussion in this section therefore treats static or quasi-static geometric structure as established at the framework level, while treating genuinely nonlinear dynamical evolution as an open problem deferred to Section B.4.

At radii approaching the displacement scale d — equivalently, where weak-field expansions in Σ cease to be accurate — the cumulative approximation $\Sigma(r) = d^2/(2r)$ is no longer the appropriate primary descriptor. Here Σ remains the accumulated scalar response, whereas the constitutive state of the medium is more naturally described by Q_{occ}; the nonlinear coefficients discussed in Section B.4 should therefore be understood as functions of the latter even when the propagated field variable is written as Σ. The physical displacement factor $D(r, d)$, however, remains finite and well defined at every finite radius: it approaches d/r at small radii, becoming large but never singular (or, with the regularization parameter ϵ, saturating at d/ϵ). In that sense, the framework does not lose its geometric content in the strong field. What fails is not the existence of a displacement geometry, but the sufficiency of the linearized scalar field law used to describe weak-field propagation.

The strong-field picture in this framework is therefore not a singularity but an extreme deformation of the transport structure. Tangential transport is expanded by large D, radial transport is compressed by $1/D$, and the net volume per cell grows with displacement. The weak-field timing proxy $1/(1 + \Sigma)$ should not itself be over-read as a completed strong-field law, but it correctly signals the limiting behavior already implied by the geometry: processes become extremely slow relative to distant coordinate time as cumulative displacement grows. There is no point at which the transport structure is required to tear or terminate. The LC network remains continuous, though severely deformed.

In this picture, a black-hole-like regime is the regime in which displacement becomes extreme over a macroscopic region. That structural interpretation follows from the nonperturbative equilibrium geometry itself. The still-open question is not whether such a regime can be

represented geometrically, but how the loaded medium dynamically reaches and sustains it under the full nonlinear law. The mass may be concentrated on or near a displacement surface rather than at a central singular point; the interior may be nearly frozen in coordinate time yet remain physically connected; and effective horizons may still arise for classes of events and observers. Those statements should be read as consequences of the established displacement geometry together with provisional strong-field interpretation, not yet as quantitatively closed predictions of the full dynamical theory.

B.4 The nonlinear wave equation

Section 8.3 suggests a cleaner way to state the strong-field dynamical problem. The weak-field scalar law is already established in terms of the cumulative displacement field Σ, but the deeper constitutive state of the medium is more naturally represented by the local occupancy burden Q_{occ}. If sustained LC excitation loads the substrate locally, then the nonlinear strong-field equation should be written not merely as a wave equation with coefficients depending on the already-accumulated field value, but as a medium-response law whose coefficients depend on the substrate loading state that is doing the sourcing.

A corresponding constitutive form is

$$\frac{1}{c_{\mathrm{eff}}^2(Q_{\mathrm{occ}})}\,\partial_t^2\Sigma - \nabla\cdot\big[\alpha(Q_{\mathrm{occ}})\,\nabla\Sigma\big] = S[Q_{\mathrm{occ}}],$$

where $c_{\mathrm{eff}}(Q_{\mathrm{occ}})$ and $\alpha(Q_{\mathrm{occ}})$ encode the loading-dependent stiffness of the medium and $S[Q_{\mathrm{occ}}]$ is the constitutive scalar source. In the weak field, $Q_{\mathrm{occ}} \ll 1$, the coefficients reduce to their background values,

$$Q_{\mathrm{occ}} \ll 1, \qquad c_{\mathrm{eff}} \to c, \qquad \alpha \to 1, \qquad S[Q_{\mathrm{occ}}] \to \frac{4\pi G}{c^2}\,\rho_{\mathrm{LC}},$$

recovering the linear wave equation of Section 5.2. In the strong field, the occupancy burden modifies the effective stiffness of the substrate and therefore the propagation law itself. The key point is that the nonlinearity is not an arbitrary correction added after the fact; it is the expected response of an already-loaded medium.

This reframing does not yet determine the explicit forms of $c_{\mathrm{eff}}(Q_{\mathrm{occ}})$, $\alpha(Q_{\mathrm{occ}})$, or $S[Q_{\mathrm{occ}}]$. Those remain part of the closure problem. But it narrows that problem. The open task is not simply to guess nonlinear functions of Σ; it is to derive how sustained excitation closure

produces the occupancy burden Q_{occ}, how that burden alters substrate response, and how the established equilibrium geometry $D(r, d)$ emerges as the stationary solution of the resulting common law.

B.5 Source-sphere enlargement and binding energy

The preceding subsections describe how the displacement field responds nonlinearly to the medium state. A further effect acts on the displacement source itself. Each mass inserts a displacement sphere of volume $\frac{4}{3}\pi d^3$ into the transport structure. When that insertion occurs in a region already displaced by another source, the transport into which the sphere is inserted is already expanded. The source sphere — the physical volume of displaced transport structure — is therefore slightly larger in coordinate terms than it would be in the undisplaced background. The displacement source geometry itself responds to the environment.

This effect is significant only when two displacement sources are close enough that the ambient D at the location of one source, due to the other, is appreciably above unity. At inter-atomic separations (angstroms), the displacement per nucleus is femtometer-scale and the effect is negligible ($d^2/r^2 \sim 10^{-10}$). At sub-nuclear separations — where the internal constituents of a nucleus interact — the displacement scales and the separations are both femtometer-scale, and the source-sphere enlargement can be significant.

A distinct but related effect occurs when two displacement spheres are close enough to merge. Two isolated sources each displace volume $\frac{4}{3}\pi d^3$; the total is twice that. When the sources merge or overlap, the combined displacement volume is less than the sum of the two individual volumes — some displaced space is shared rather than independently sourced. Since the gravitational mass of a system is set by its total displacement through $d^2 = 2Gm/c^2$, a reduction in total displacement volume corresponds to a reduction in mass. The mass deficit is the binding energy.

The source-sphere enlargement and the volume merging act in opposite directions. Enlargement increases each source's effective displacement in the other's environment; merging reduces the total displacement volume when the sources combine. Binding energy — the net mass deficit of a composite system — is the result of the volume reduction from merging exceeding the enlargement from mutual displacement. At the sub-nuclear scale, where nuclear binding energies are of order 0.9% of the rest mass, both effects are significant and their net balance determines the observed binding curve.

Appendix C. Second Post-Newtonian Predictions

The effective coordinate geometry of Section 5.3 matches the Schwarzschild metric at first post-Newtonian (1PN) order. At second order the two diverge, producing specific predictions that differ from general relativity. This appendix records those differences and estimates their physical magnitude.

C.1 Metric expansions

The framework's exact metric components are

$$f = \frac{1}{(1+\Sigma)^2}, \qquad g = D^2 = 1 + 2\Sigma, \qquad h = 1,$$

with $\sigma = r_s/r = 2\Sigma$ as the weak-field expansion parameter. The Schwarzschild components in the same coordinates are

$$f_S = 1 - \sigma, \qquad g_S = \frac{1}{1-\sigma}, \qquad h_S = 1.$$

Expanding to $O(\sigma^2)$:

$$\begin{aligned} f &= 1 - \sigma + \tfrac{3}{4}\sigma^2 + \cdots, & f_S &= 1 - \sigma, \\ g &= 1 + \sigma, & g_S &= 1 + \sigma + \sigma^2 + \cdots, \\ fg &= 1 - \tfrac{1}{4}\sigma^2 + \cdots, & f_S g_S &= 1. \end{aligned}$$

The two metrics agree at $O(\sigma)$ in every component. At $O(\sigma^2)$ they differ in three ways.

First, the time component f acquires a $\frac{3}{4}\sigma^2$ correction in the framework because $1/(1+\Sigma)^2$ is smooth through $\Sigma = 1$, whereas Schwarzschild is exactly linear in σ with a zero at $\sigma = 1$. The framework's time dilation approaches zero asymptotically rather than reaching it at a finite radius.

Second, the radial component g is exactly linear in the framework ($D^2 = 1 + \sigma$), while Schwarzschild's $1/(1-\sigma)$ carries all powers of σ. The framework's displacement geometry truncates at first order in the radial deformation.

Third, the product fg departs from unity at $O(\sigma^2)$ in the framework, while Schwarzschild

maintains $f_S g_S = 1$ exactly. This is a structural difference: in Schwarzschild, the radial light-speed coordinate factor $\sqrt{f/g}$ is exactly $1-\sigma$; in the framework it is $1-\sigma+\frac{7}{8}\sigma^2+\cdots$.

C.2 Orbit equation at 2PN

The orbit equation for a test mass on the framework's effective geometry, expanded to $O(r_s^2)$, takes the form

$$\frac{d^2u}{d\phi^2}+u=\frac{GM}{L^2}+\frac{3GM}{c^2}\,u^2+r_s^2\left(\frac{E^2}{4L^2c^2}\,u-2u^3-\frac{u}{L^2}\right)+O(r_s^3),$$

where $u=1/r$, L is the conserved angular momentum per unit mass, and E is the conserved energy per unit mass. The Schwarzschild orbit equation has no $O(r_s^2)$ correction in this form (the 2PN Schwarzschild corrections enter through the energy-angular momentum relation rather than as additional terms in the orbit equation itself).

The leading new term is $-2r_s^2u^3$. For a nearly circular orbit at radius r_0 with $u_0=1/r_0$, this correction is of order r_s^2/r_0^3 relative to the Newtonian term u_0. Its ratio to the 1PN correction is $\sigma_0=r_s/r_0$ — the same small parameter that controls the post-Newtonian expansion.

C.3 Physical magnitudes

For Mercury ($\sigma\approx 5\times10^{-8}$):

$$\frac{\text{2PN correction}}{\text{1PN precession}}\sim\sigma\approx 5\times10^{-8}.$$

The 2PN contribution to Mercury's perihelion advance is approximately 0.002 milliarcsec per century — roughly ten orders of magnitude below current measurement precision.

For the Hulse-Taylor binary ($\sigma\approx 2\times10^{-6}$):

$$\frac{\text{2PN correction}}{\text{1PN precession}}\sim 2\times10^{-6}.$$

With a 1PN precession of order 4 degrees per year, the 2PN correction is approximately 0.03 arcsec per year. Current timing precision for the Hulse-Taylor system constrains the periastron advance to within 0.04% of the GR prediction, which corresponds to roughly 5 arcsec per year — well above the predicted 0.03 arcsec/yr difference. The 2PN divergence is therefore marginally below current sensitivity for compact binaries but could become accessible with

improved pulsar timing.

C.4 Light propagation at second order

The coordinate radial speed of light is $c\sqrt{f/g}$. In the framework this expands as

$$c\sqrt{f/g} = c\big(1 - \sigma + \tfrac{7}{8}\sigma^2 + \cdots\big)\,,$$

while in Schwarzschild the exact result is $c(1-\sigma)$. The second-order correction $\frac{7}{8}\sigma^2$ modifies the integrated time delay and deflection angle at the same post-Newtonian order. For Shapiro delay, this would produce a correction to the logarithmic time-delay formula proportional to $\sigma^2 \ln(4L_1L_2/b^2)$. For light deflection, the correction modifies the bending angle at $O(GM/(c^2b))^2$.

The physical magnitude for solar-system measurements is again small: $\sigma_\odot^2 \sim 10^{-12}$ at the solar limb. Precision ranging experiments or high-accuracy astrometric missions operating at the micro-arcsecond level could in principle access this regime, but it remains beyond current observational reach.

C.5 Status

The 2PN divergence between the displacement framework and general relativity is a genuine prediction, not a tunable parameter. It follows directly from the exact forms of the metric components, which are themselves determined by the framework's internal logic: $f = 1/(1+\Sigma)^2$ from the clock-rate relation, $g = D^2$ from the radial deformation, and $h = 1$ from the coordinate anchoring of $F \cdot H$. The prediction is that the Schwarzschild horizon is replaced by asymptotic time dilation, and the radial deformation is exactly linear in σ. Both features produce specific second-order corrections that are currently below observational thresholds but are in principle testable.

Appendix D. Mass as Surface Area and the Displacement Composition Rule

The relation $d^2 = 2Gm/c^2$ has a direct geometric reading that determines how masses compose in the displacement framework.

D.1 Surface area is mass

The surface area of a displacement sphere of radius d is

$$S = 4\pi d^2 = \frac{8\pi G m}{c^2}.$$

Surface area is exactly linear in mass. This is not a dimensional accident — it is the content of the relation $d^2 \propto m$. The displacement sphere's surface area is proportional to the mass that sources it, with the proportionality constant set by G and c.

An immediate consequence is that surface areas add linearly. For N displacement spheres with individual surface areas $S_n = 4\pi d_n^2$, the total surface area is $N\, S_n$. The combined displacement parameter is $d^2_{\text{combined}} = N\, d_n^2$, and the combined mass is

$$m_{\text{combined}} = \frac{d^2_{\text{combined}}\, c^2}{2G} = N\, m_n.$$

Masses add exactly when surface areas add. This is why Newtonian gravity is linear in mass: the linearity is the surface-area composition of displacement spheres.

D.2 Binding energy as shared surface area

When two displacement spheres overlap, the contact region reduces the total exposed surface area. The total surface area of the combined system is less than the sum of the individual surface areas by the area of the shared patch. Since mass is proportional to surface area, the mass deficit is proportional to the shared area:

$$\frac{\Delta m}{m} = \frac{\Delta S}{S}.$$

The fractional mass deficit equals the fractional surface area deficit, one to one. No geometric correction factor is needed.

For iron, the nuclear binding energy is approximately 8.8 MeV per nucleon out of 939 MeV rest mass, a fractional deficit of about 0.94%. The implied surface area sharing is therefore also about 0.94% per nucleon. The corresponding contact patch radius is approximately 20% of the nucleon displacement radius — geometrically consistent with close-packed displacement spheres barely overlapping at their boundaries.

D.3 Volume addition and its consequences

If displacement volumes were the additive quantity rather than surface areas, the composition rule would be $d^3_{\text{combined}} = N\, d_n^3$, giving $d_{\text{combined}} = N^{1/3}\, d_n$ and $m_{\text{combined}} = N^{2/3}\, m_n$. This produces massive overbinding — a 74% mass deficit for 56 nucleons, compared to the observed 0.94%. Volume addition is therefore not the composition rule for displacement in the weak-overlap regime.

However, volume becomes relevant in the strong-overlap regime. When two displacement spheres fully merge, the combined volume is the sum. The transition from surface-area-additive (weak overlap, small binding) to volume-additive (full merger, large binding) spans the range between nuclear physics and black hole coalescence.

D.4 Connection to black hole mergers

The working notes record that LIGO merger remnant masses are best fit by $M_A^n + M_B^n = M_C^n$ with $n \approx 1.1$. In the displacement picture, the exponent n characterizes the composition rule:

$$\begin{aligned} n = 1: &\quad \text{surface areas add (exact mass additivity, no deficit)}, \\ n = \tfrac{2}{3}: &\quad \text{volumes add (maximum binding, } 20.6\% \text{ deficit for equal masses)}. \end{aligned}$$

The LIGO value $n \approx 1.1$ is close to surface-area addition with a small correction. For equal-mass mergers, it gives a mass deficit of approximately 6% — the energy radiated as gravitational waves during the merger. The coalescence is mostly surface-area-additive, with a volume-interaction correction at the merger interface where the displacement spheres interpenetrate.

The exponent n therefore encodes the degree of displacement-sphere interpenetration during the merger. A value near 1 means the spheres join with minimal volume sharing; a value near $2/3$ would mean complete volumetric merger. The observed $n \approx 1.1$ indicates that even in the extreme strong-field regime of black hole coalescence, the composition is predominantly surface-area-like.

Appendix E. Rotating-Source Effective Metric

The static effective geometry of Section 5.3 describes a non-rotating source. This appendix

extends that geometry to slowly rotating sources by developing the vector sector of the retarded displacement integral.

E.1 The retarded vector integral

Just as the scalar cumulative displacement Σ is sourced by the retarded mass density, the vector displacement is sourced by the retarded mass current. By direct analogy with Section 5.2, the vector component of the displacement field is

$$A_i(\mathbf{x}, t) = \frac{G}{c^3} \int \frac{\rho_{\mathrm{LC}}(\mathbf{x}', t_r)\, v_i(\mathbf{x}', t_r)}{|\mathbf{x} - \mathbf{x}'|}\, d^3x',$$

where v_i is the velocity of the source element. This integral has exactly the structure of the electromagnetic vector potential, with G/c^3 replacing $1/(4\pi\varepsilon_0 c)$ and mass current replacing charge current.

For a slowly rotating source with angular momentum $J = I\omega$, the leading multipole of this integral is the magnetic-dipole-type term:

$$A_\phi \sim \frac{GJ \sin^2\theta}{c^3\, r}.$$

This is the gravitomagnetic vector potential. Its curl gives the gravitomagnetic field, which is the transport twist developed in Sections 6 and 7.

E.2 The slow-rotation effective metric

The vector displacement enters the effective coordinate metric as a cross term between the time coordinate and the azimuthal angle. Combined with the scalar sector from Section 5.3, the full slow-rotation effective metric is

$$ds^2 = -\frac{c^2}{(1+\Sigma)^2}\, dt^2 + D^2\, dr^2 + r^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big) + \frac{4GJ\sin^2\theta}{c^2\, r}\, dt\, d\phi.$$

The scalar sector (f, g, $h = 1$) is unchanged by the rotation at leading order. The rotation enters only through the cross term $g_{t\phi}$, which is linear in J, odd under reversal of spin, and vanishes on the rotation axis ($\theta = 0, \pi$).

This is the same structure as the slow-rotation limit of the Kerr metric in Boyer-Lindquist

coordinates. At leading order in both Σ and J, the two metrics agree term by term. The scalar sector matches at 1PN (as established in Section 5.4 and Appendix C), and the vector sector matches at the leading gravitomagnetic order.

E.3 Frame-drag angular velocity and gyroscope precession

The frame-dragging angular velocity — the rate at which the local transport structure is twisted by the source rotation — is

$$\omega_{\text{drag}} = \frac{g_{t\phi}}{g_{\phi\phi}} = \frac{4GJ}{c^2\, r^3}.$$

This is independent of latitude at leading order and falls as $1/r^3$, the characteristic dipole scaling of the gravitomagnetic field.

For a gyroscope on a polar orbit, the gyroscope's spin axis is carried by this twisted transport [7]. The orbit-averaged precession rate depends on the angular projection of the twist onto the orbital geometry. For the Earth, using the thin-shell angular momentum $J_{\text{shell}} = \frac{2}{3} M R^2 \omega$ and the GPB orbital radius, the thin-shell estimate is approximately 82 mas/yr. Applying the Earth moment-of-inertia correction $I_\oplus/(MR^2)/(2/3) \approx 0.496$ yields approximately 41 mas/yr, close to the observed 39.2 mas/yr. The $\sim 4\%$ residual may reflect either the specific polar-orbit angular projection or a genuine higher-order difference between the framework and general relativity in the vector sector.

E.4 Rapidly rotating sources and the equilibrium surface

The metric of Section E.2 is the displacement framework's analog of the slow-rotation Kerr metric. The structural correspondence is exact at leading order in both the scalar and vector sectors. The scalar sector diverges from Schwarzschild at second post-Newtonian order (Appendix C); whether an analogous second-order divergence exists in the vector sector remains to be computed.

Beyond the slow-rotation limit, the displacement framework admits a nonperturbative treatment of the rotating equilibrium. A self-gravitating spinning body in displacement equilibrium must satisfy equal realized time dilation at every latitude on its surface. Gravitational time dilation (from the cumulative displacement Σ) and velocity time dilation (from the surface rotation

speed) both reduce the local clock rate, and the equilibrium shape is the surface on which their combined effect is uniform:

$$\frac{c^2}{(1+\Sigma(r(\phi)))^2} - r(\phi)^2\omega^2\cos^2\phi = S_0^2,$$

where S_0 is the uniform residual spin capacity and ϕ is latitude. At the pole ($\phi = \pi/2$), the rotation velocity vanishes and the condition reduces to $S_0 = c/(1+\Sigma(r_p))$, fixing S_0 in terms of the polar radius r_p. At the equator ($\phi = 0$), the rotation velocity is maximal, and the equilibrium radius is correspondingly larger. The surface is oblate, with the oblateness determined by the spin.

In nondimensionalized form, setting $\rho = 2r/d^2$ (radius in units of the displacement scale) and $\alpha = d^2\omega/(2c)$ (dimensionless spin parameter), the equilibrium condition becomes

$$\left(\frac{1}{\rho_p} - \frac{1}{\rho}\right)\left(2 - \frac{1}{\rho} - \frac{1}{\rho_p}\right) = \alpha^2\rho^2\cos^2\phi.$$

This is exact and implicit in $\rho(\phi)$. For each latitude, it determines the equilibrium surface radius as a function of the two free parameters ρ_p (polar radius in displacement-scale units) and α (dimensionless spin). In the slow-rotation limit $\alpha \ll 1$, it linearizes to a small oblate perturbation of the spherical equilibrium, recovering the slow-rotation metric of Section E.2. For rapid rotation, the surface becomes highly oblate, with the equatorial radius potentially many times the polar radius.

The maximum spin occurs when the equatorial surface velocity reaches c, giving $\rho_e = 1/\alpha$. At this limit the equatorial matter has no residual spin capacity ($S = 0$), and no internal LC oscillation is possible at the equator. The polar matter, by contrast, retains full isotropic spin capacity reduced only by the gravitational depth. This asymmetry between directional (velocity) and isotropic (gravitational) time dilation at different latitudes is a distinctive feature of the displacement picture that has no direct analog in the Kerr metric, where the equilibrium surface geometry is algebraically determined by the Kerr parameters rather than by a physical balance of time-dilation mechanisms.

The astrophysical consequences of the rapidly rotating equilibrium — including polar mass accumulation, jet formation, and the spin-dependent equilibrium shape — are developed separately. For the present paper, the equilibrium surface equation serves as the natural strong-field, rapidly-rotating extension of the slow-rotation effective metric, completing the

framework's analog of the Kerr solution.

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